Supplementary Material for Protein Complex Identification via Joint Embedding Learning with a Variational Graph Autoencoder

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1. Preprocessing Edge Statistics

To make the preprocessing stage more transparent, we report the edge statistics before and after temporal and spatial filtering in Table S1. For the three yeast benchmarks, temporal filtering is first applied using gene-expression information, followed by spatial filtering based on subcellular localization consistency. For the Homo sapiens benchmark, temporal filtering is not applied because the benchmark is constructed from STRING physical interactions; therefore, we first retain high-confidence undirected STRING interactions with combined score greater than 700 and then apply GO cellular-component (CC) spatial filtering.

2. Analysis of GO Attribute Embeddings

To examine whether the learned GO attribute embeddings reflect ontology-derived GO structure, we performed an additional post hoc analysis on the retained GO terms used in Biogrid datasets. After excluding terms without valid ontology statistics in the current preprocessing pipeline, 135 GO terms remained, yielding 5,301 GO term pairs for pairwise analysis. We compared GO DAG distance with latent-space distance, and GO semantic similarity with embedding cosine similarity. The results show modest but non-random alignment between the learned embedding geometry and ontology-derived GO relations. In particular, the Spearman correlation between GO DAG distance and embedding distance is 0.0665, and the Spearman correlation between Resnik semantic similarity and embedding cosine similarity is 0.0638. These findings suggest that the shared latent space captures limited semantic organization among GO terms.

3. Effect of Graph Reweighting

To examine whether graph reweighting improves intra-complex cohesion, we compare the learned weights of observed PPI edges inside gold-standard complexes with those of background observed PPI edges. An edge is regarded as an intra-complex edge if its two endpoint proteins appear in at least one common gold-standard complex; all remaining observed PPI edges are treated as background edges. As shown in Table S3, intra-complex edges consistently receive higher average weights than background edges across all evaluated datasets. This suggests that graph reweighting does not merely preserve the preprocessed topology, but further differentiates retained interactions according to learned structural and GO-derived functional consistency.

4. Runtime Analysis

We further report the runtime and model-size statistics of VAJEL on the evaluated benchmarks. The GPU training time corresponds to the full training procedure over 200 epochs on a single NVIDIA GeForce RTX 3080 GPU with 10GB memory. We also report the number of trainable parameters and the estimated memory footprint of the dominant dense reconstruction tensors.

In the current implementation, the sparse graph convolutional encoder scales with the number of observed PPI edges, whereas the VGAE decoder reconstructs the adjacency matrix using dense pairwise scores. The trainable parameter count is determined by the dataset size because one attribute-side dense layer takes the number of proteins as its input dimension. The dominant decoder memory comes from the $N \times N$ adjacency reconstruction and the $N \times A$ protein–attribute reconstruction tensors, where N is the number of proteins and A is the number of GO attributes. Therefore, the dense reconstruction stage has $O(N^2 + NA)$ memory cost. The reported dense reconstruction memory does not include gradients, optimizer states, temporary tensors, or TensorFlow runtime overhead; thus the actual peak memory during training is higher.

The downstream complex-identification stage has a different bottleneck. Edge reweighting is computed only for observed PPI edges, avoiding all-pairs similarity computation during this step. However, clique mining and core-attachment expansion can become expensive on dense graphs or graphs with many maximal cliques. Thus, VAJEL is practical for sparse PPI networks with several thousand proteins, as in the yeast benchmarks and the evaluated Homo sapiens benchmark, but may become impractical for very dense PPI networks, large cancer-specific networks, or networks with tens of thousands of proteins unless approximation strategies are used.

5. Additional analysis of unmatched complexes

To complement the main-text case study, we provide representative unmatched complexes and VAEPC1-specific pathway analysis in this appendix. Table S5 lists representative unmatched complexes identified by VAJEL, together with their LAGO significance p-values and top associated GO terms. VAEPC1 corresponds to the first entry in this table. Table S6 further reports the KEGG pathway enrichment

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Table S1
Edge counts before and after preprocessing. For yeast datasets, spatial filtering is applied after temporal filtering; for Homo sapiens, only CC-based spatial filtering is applied to high-confidence STRING physical interactions.

Dataset	Initial	After temporal	Temporal removed	After spatial
BioGRID	59,748	43,161	16,587 (27.76%)	35,294
DIP	17,201	13,142	4,059 (23.60%)	10,330
Krogan14K	14,076	10,146	3,930 (27.92%)	7,878
Homo sapiens	86,355	N/A	N/A	74,700

Table S2
Alignment between learned GO attribute embeddings and ontology-derived GO relations on Biogrid benchmarks.

Statistic	Value
Retained GO terms used in the analysis	135
GO term pairs used in the analysis	5,301
Spearman correlation between GO DAG distance and embedding distance	0.0665
Spearman correlation between Resnik similarity and embedding cosine similarity	0.0638

results for VAEPC1. In addition to VAEPC1, the remaining unmatched complexes also show significant functional enrichment. VAEPC2 is associated with mRNA splicing via spliceosome, VAEPC3 and VAEPC4 are enriched in spliceosomal complex-related terms, VAEPC5 is associated with cytosolic ribosome, and VAEPC6 is again enriched in preribosome large-subunit precursor-related functions. These results suggest that the unmatched complexes are not random protein groups, although their unmatched status should be interpreted relative to the selected gold-standard coverage and overlap threshold. Complete machine-readable results for all unmatched complexes, including dataset-level CSV files, are provided in the supplementary online material and mirrored in our public GitHub repository.

Table S3
Effect of graph reweighting on intra-complex cohesion. Intra edges denote observed PPI edges whose endpoints appear in the same gold-standard complex; background edges denote the remaining observed PPI edges.

Dataset	Obs. edges	Intra edges	Bg. edges	Mean intra weight	Mean bg. weight
BioGRID	35,294	9,487	25,807	0.6789	0.2730
DIP	10,330	3,098	7,232	0.7063	0.2927
Krogan14K	7,878	2,801	5,077	0.7133	0.2574
Homo sapiens	74,700	19,455	55,245	0.8602	0.7877

Table S4
Runtime and model-size statistics of VAJEL.

Dataset	#Proteins	#Edges	#GO attr.	Trainable parameters	GPU training time (s)	Dense recon. memory (MiB)
BioGRID	4,696	35,294	142	342,720	292.348	86.67
DIP	3,739	10,330	142	281,472	163.130	55.36
Krogan14K	2,670	7,878	142	213,056	88.286	28.64
Homo sapiens	10,051	74,700	14,036	1,574,656	3799.226	923.53

Table S5
Representative unmatched protein complexes identified by VAJEL, with LAGO significance p-values and top associated GO terms.

Index	Unmatched complex	Minimum p-value	GO ID	GO term
1	YLL034C YGL111W YOL077C YNL061W YOR272W YER006W YHR088W YDL031W YFL002C YDR496C YNL110C YKR081C YPL093W YNL002C YGR103W YLR002C YMR290C YLL008W YPL211W YMR049C YPL198W	5.763e-40	GO:0030687	preribosome, large subunit precursor
2	YLR275W YLR117C YPR182W YKL173W YMR125W YLR424W YJR022W YMR213W YGL128C YER112W YML049C YGR013W YER146W YDL030W YPL151C YNR011C YBR152W YPR178W YBL074C YGR074W	4.134e-34	GO:0000398	mRNA splicing, via spliceosome
3	YIL061C YPL151C YNR011C YML049C YGR074W YFL017W-A YGR091W YMR125W YPR101W YKL173W YPR178W YMR213W YDL030W YMR288W YJL203W YDR473C YLR424W YLR117C	7.153e-33	GO:0005681	spliceosomal complex
4	YPR178W YPR101W YPL190C YLR424W YDL030W YMR288W YER172C YNR011C YER112W YLR117C YPR182W YAL032C YMR125W YML049C YBR119W YLR298C YLR275W YMR213W YDR473C YPL178W	2.718e-33	GO:0005681	spliceosomal complex
5	YLR448W YNL096C YKL180W YGR118W YLR029C YNL255C YBR031W YJL177W YMR242C YDR471W YGL147C YML063W YMR194W YHL001W YNL301C YBL092W YBL027W YDL082W YOL127W YFR032C-A	1.323e-28	GO:0022626	cytosolic ribosome
6	YPL093W YNL110C YKR081C YNL061W YLL008W YMR049C YPL043W YMR290C YDL213C YDR496C YGL111W YNL002C YKL172W YBR142W YHR066W YOR272W YGR103W YPR016C	2.111e-27	GO:0030687	preribosome, large subunit precursor

Table S6
KEGG pathway enrichment analysis of unmatched protein complex VAEPC1.

Pathway	Candidate genes with pathway annotation (5)	All genes with pathway annotation (2,124)	p-value	Pathway ID
Ribosome biogenesis in eukaryotes	3 (60%)	74 (3.48%)	3.860e-4	ko03008